

AETHEXER / EXITZ — Project Bible

Universal Truth Layer & Carbon Displacement Token

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Purpose of This Document: This is a self-contained reference document for Claude AI sessions. Paste this at the beginning of any conversation to give Claude full architectural context on the Aethexer/EXITZ system. This covers both the protocol layer (within Bedrock ESG codebase) and the front-facing landing page (aethexer.com, built by Claude separately).

1. System Identity & Mission

Aethexer is the **Universal Truth Layer** — a proprietary protocol built by **ZerO** that provides military-grade security, environmental verification, and carbon displacement tracking for digital commerce and supply chain operations.

EXITZ is the native token of the Aethexer protocol — a Solana Token-2022 asset representing verified carbon displacement credits generated through the digitization of physical transport and logistics processes.

The Aethex Protocol is the intellectual property of ZerO and represents the cryptographic backbone of the Bedrock ESG platform.

2. Architecture Overview

Two Codebases, One System

Component	Location	Built By
Aethex Protocol Engine	Inside Bedrock ESG codebase (/home/ubuntu/ scrapnet_dashboard/nex- tjs_space/)	Abacus AI Agent
aethexer.com Landing Page	Separate project/conversation	Claude AI

Critical Integration: Both share the same Solana identity (Aethexer Sentinel), same database, and the protocol engine routes live inside the Bedrock ESG Next.js application.

3. The Aethex Protocol — Three-Tier Security Model

The protocol provides “**Armored Digital Packets**” — cryptographically secured data containers that replace physical armored transport, generating measurable carbon displacement.

Tier 1: Ballistic Layer

- **Encryption:** AES-GCM-256
- **Purpose:** Payload-level encryption of all packet contents
- **Implementation:** `encryptPayload()` in `lib/aethex-protocol.ts`
- **IV:** 12-byte random initialization vector per packet
- **Auth Tag:** 16-byte authentication tag for tamper detection

Tier 2: Guardian Layer

- **Mechanism:** 2-of-2 Multi-Signature Verification
- **Purpose:** Dual-party authorization for packet operations
- **Signers:** Sender + Receiver must both sign
- **Implementation:** `createGuardianSignature()`, `verifyGuardianConsensus()`

Tier 3: Stealth Layer

- **Mechanism:** Zero-Knowledge Proof (ZKP)
- **Purpose:** Verify packet integrity without revealing contents
- **Implementation:** `generateStealthProof()`, `verifyStealthProof()`
- **Output:** Proof hash that validates data existence without exposure

Packet Lifecycle

- | | | |
|---------------|---|--|
| 1. CREATED | → | Payload encrypted (Ballistic) |
| 2. SIGNED | → | Multi-sig applied (Guardian) |
| 3. VERIFIED | → | ZKP generated (Stealth) |
| 4. IN_TRANSIT | → | Packet moving between parties |
| 5. DELIVERED | → | Received and verified |
| 6. SETTLED | → | Carbon displacement calculated and recorded |

4. EXITZ Token

Token Specification

Property	Value
Standard	SPL Token-2022
Network	Solana Devnet
Mint Address	44SrHT9Qwyz2jiiJkRF1zHKF2NFTgTC8m6ubPT-b5qJBJ
Decimals	0 (whole units only)
Mint Authority	Aethexer Sentinel (9tsq8qB4P9uPRSHr-jXLaDQJd93nuo4wEEtkR1tAJRAQ3)
Freeze Authority	Aethexer Sentinel

Token Metadata

```
{
  "name": "EXITZ",
  "symbol": "EXITZ",
  "uri": "",
  "additionalMetadata": [
    ["backtrace_id", "BT-C9C4C5"],
    ["engine", "Bedrock ESG"],
    ["protocol", "Aethex Universal Truth Layer"],
    ["standard", "ISO 14064-2"]
  ]
}
```

Genesis Emission

- **Amount:** 1.0 EXITZ
- **Recipient:** Aethexer Sentinel wallet
- **Script:** scripts/genesis-emission.ts
- **Status:** Confirmed on Solana Devnet
- **Purpose:** Protocol initialization / proof of capability

Token Creation Script

- **File:** scripts/mint-exitz.ts
- **Method:** Uses @solana/spl-token Token-2022 extensions
- **Extensions:** TokenMetadata (on-chain metadata without Metaplex)
- **Flow:** createMint() → createInitializeMetadataPointerInstruction() → createInitializeInstruction() → createUpdateFieldInstruction()

5. Trade Settlement & GHG Displacement Engine

File: lib/aethex-engine.ts

Core Constants

```
DISPLACEMENT_FACTOR_KG_CO2E_PER_MILE = 2.7 // kg CO2e displaced per digital mile
SETTLEMENT_FEE_PERCENT = 0.5 // 0.5% settlement fee
MIN_PACKET_VALUE_USD = 100 // Minimum packet value
MAX_BATCH_SIZE = 1000 // Maximum batch transfer size
```

Current State (as of latest data)

- **Total Packets Processed:** 779
- **Total CO2e Displaced:** ~6.7 tonnes
- **Settlement Status:** Active

GHG Displacement Calculation

$$\text{Displacement_CO2e} = \text{packets_count} \times \text{avg_distance_miles} \times 2.7 \text{ kg CO2e/mile}$$

The engine calculates how much CO2e would have been emitted by physical armored transport (trucks, security vehicles) and credits that amount as displacement when the equivalent operation is performed digitally via Armored Digital Packets.

Settlement API (/api/aethex-settlement)

GET Operations (via ?action= parameter)

Action	Description
status	Current engine status and statistics
baselines	Environmental baselines and thresholds
settlements	Settlement history records
calculate	Calculate displacement for given parameters

POST Operations (via action in body)

Action	Description
transfer-packets	Transfer Armored Digital Packets between parties
batch-transfer	Batch transfer multiple packets
verify-packet	Verify packet integrity and authenticity
get-current-state	Get full current protocol state

6. Armored Digital Packets

File: `lib/aethex-protocol.ts`

Packet Structure

```
interface ArmoredDigitalPacket {  
  id: string; // Unique packet identifier  
  payload: EncryptedPayload; // AES-GCM-256 encrypted data  
  guardian: GuardianSignature; // 2-of-2 multi-sig  
  stealth: StealthProof; // Zero-knowledge proof  
  metadata: PacketMetadata; // Timestamps, route info  
  displacement: DisplacementRecord; // CO2e displacement data  
}
```

Digital Packet API (`/api/aethex/digital-packet`)

GET Operations

Action	Description
<code>list</code>	List all packets with optional filters
<code>verify</code>	Verify a specific packet by ID
<code>stats</code>	Packet statistics and totals

POST Operations

Action	Description
<code>create</code>	Create new Armored Digital Packet
<code>verify</code>	Verify packet integrity
<code>calculate-savings</code>	Calculate CO2e savings for a route

7. Retroactive Harvest System

File: `lib/aethex-protocol.ts` (RetroactiveHarvest class) + `/api/aethex/retroactive-harvest`

Purpose

Ingests historical enterprise resource planning (ERP) data and retroactively generates environmental impact assessments and Material Passports.

Supported ERP Systems

- **SAP** (S/4HANA)
- **Oracle** (Cloud ERP)

- **Viewpoint** (Vista / Spectrum, construction-specific)

Compliance Standards

- **2026 Caltrans Environmental Standards** — California Department of Transportation requirements
- **ISO 14064-2** — GHG quantification at project level
- **ISO 14040** — Life Cycle Assessment

Harvest Workflow

1. INGEST

2. AUDIT

3. CALCULATE

4. MINT

5. CERTIFY

➔

➔

➔

➔

➔

Import ERP **data** (procurement, **logistics**, materials)

Verify against 2026 Caltrans standards

Compute retroactive environmental impact

Generate Material Passports with Solana attestation

Issue compliance certificates

Material Passports

Digital documents that travel with physical materials, containing:

- Full chain-of-custody
- Environmental impact data (embodied carbon, transport emissions)
- Compliance status against applicable standards
- Solana attestation proof (backtrace: `BT-C9C4C5`)

Retroactive Harvest API (`/api/aethex/retroactive-harvest`)

GET Operations

Action	Description
<code>status</code>	Harvest system status
<code>passports</code>	List generated Material Passports
<code>compliance</code>	Compliance audit results

POST Operations

Action	Description
<code>ingest</code>	Ingest ERP data for processing
<code>audit</code>	Run compliance audit against standards
<code>mint-passport</code>	Generate and attest Material Passport

8. Integration with Bedrock ESG

Shared Infrastructure

Resource	Shared?	Details
Database	✓ Yes	Same PostgreSQL instance, same Prisma schema
Solana Identity	✓ Yes	Same Aethexer Sentinel wallet
API Routes	✓ Yes	Aethex routes live inside Bedrock ESG codebase
Auth System	✓ Yes	Same NextAuth.js session/JWT
LLM API	✓ Yes	Same Abacus AI RouteLLM access
File Storage	✓ Yes	Same AWS S3 bucket

Data Flow

```

Bedrock ESG Modules → Universal Ledger → Aethex Settlement Engine
                        ↓
                    EXITZ Token Minting
                        ↓
                    Solana Attestation

```

The Aethex engine consumes environmental data from all Bedrock ESG modules (asphalt calculations, biochar CDR, metal recovery) and translates verified environmental impact into EXITZ token issuance.

9. Key Files Reference

Protocol Core

```

lib/aethex-protocol.ts    # Full Aethex Protocol implementation
                          # - AethexProtocol class
                          # - Ballistic/Guardian/Stealth layers
                          # - Armored Digital Packets
                          # - Retroactive Harvest system

lib/aethex-engine.ts      # Trade Settlement & GHG Displacement Engine
                          # - Displacement calculations
                          # - Settlement fee processing
                          # - Batch transfer logic

```

API Routes

```
app/api/aethex-settlement/route.ts      # Main settlement endpoint
app/api/aethex/digital-packet/route.ts  # Armored Digital Packet CRUD
app/api/aethex/retroactive-harvest/route.ts # ERP ingestion & Material Passports
```

Token Scripts

```
scripts/mint-exitz.ts      # EXITZ token creation (Token-2022)
scripts/genesis-emission.ts # Genesis 1.0 EXITZ mint
```

UI Components

```
components/aethex-protocol-widget.tsx # Protocol status widget
components/solana-provider.tsx        # Wallet adapter (shared with Bedrock)
```

10. aethexer.com — Front Landing Page

Important Context

- **Built by:** Claude AI in a **separate conversation/project**
- **NOT in this codebase** — the landing page source code lives in a different Abacus AI Agent conversation
- **Purpose:** Public-facing marketing/information site for the Aethex Protocol
- **Relationship:** Points to and references the protocol engine running on bedrockesg.com

When Working on aethexer.com

If you're in a Claude session for the aethexer.com landing page:

- The protocol logic is NOT in your codebase — it lives in Bedrock ESG
- API calls should target `bedrockesg.com/api/aethex-*` endpoints
- The landing page is presentational; the engine runs on Bedrock ESG
- Use the Bedrock ESG Project Bible for understanding the backend

When Working on Bedrock ESG

If you're modifying Aethex Protocol routes in Bedrock ESG:

- Changes to `/api/aethex-*` routes affect both bedrockesg.com and aethexer.com
- The aethexer.com landing page may reference these endpoints
- Ensure backward compatibility when modifying API contracts

11. Key Constants & Formulas

Displacement Factor

```
2.7 kg CO2e per mile = physical armored transport displacement factor
Source: Calculated from average armored vehicle fuel consumption + security escort overhead
```

Settlement Fee

0.5% of transaction value
Minimum packet value: \$100 USD
Maximum batch size: 1,000 packets

Security Parameters

AES-GCM Key Size: 256 bits
IV Length: 12 bytes
Auth Tag Length: 16 bytes
Multi-Sig Required: 2 of 2
ZKP Hash: SHA-256 based proof

Token Economics

EXITZ Supply: Elastic (minted per verified displacement)
Decimals: 0 (whole units)
Mint Authority: Aethexer Sentinel only
Burn Mechanism: Retirement via Universal Ledger

12. Terminology Glossary

Term	Definition
Aethexer	The protocol system name — “Universal Truth Layer”
EXITZ	Native carbon displacement token (Solana Token-2022)
Aethexer Sentinel (Node-01)	The institutional Solana wallet identity used for all platform operations (formerly Kato Service Identity)
Armored Digital Packet	Cryptographically secured data container replacing physical transport
Ballistic Layer	AES-GCM-256 encryption tier
Guardian Layer	2-of-2 multi-signature verification tier
Stealth Layer	Zero-Knowledge Proof verification tier
Retroactive Harvest	System for processing historical ERP data into environmental credits
Material Passport	Digital document tracking material chain-of-custody and environmental impact
Backtrace	Audit trail identifier (BT-C9C4C5) linking all operations
ZerO	The entity/organization behind both Bedrock ESG and Aethexer
Sovereign Notary	Solana Memo-based attestation system
HUM Listener	Energy grid baseline monitoring service
Universal Ledger	Master environmental accounting ledger in Bedrock ESG
Displacement Factor	2.7 kg CO2e/mile — carbon savings from digital vs. physical transport

13. Cross-Reference Guide

When working across both projects, here’s how they connect:

aethexer.com (Landing Page)		bedrockesg.com (Engine)	
Marketing / Public Info	↔	Protocol Implementation	
UI/UX / Brand Identity	→	API Endpoints (/api/aethex-*)	
	←	Data / Statistics	
	←	Solana Attestation Proofs	
	←	EXITZ Token State	

Shared Identity:

- Same Solana wallet (Aethexer Sentinel)
- Same backtrace ID (BT-C9C4C5)
- Same EXITZ token mint address
- Same ZerO organizational branding
